

Toxicity Classification Report

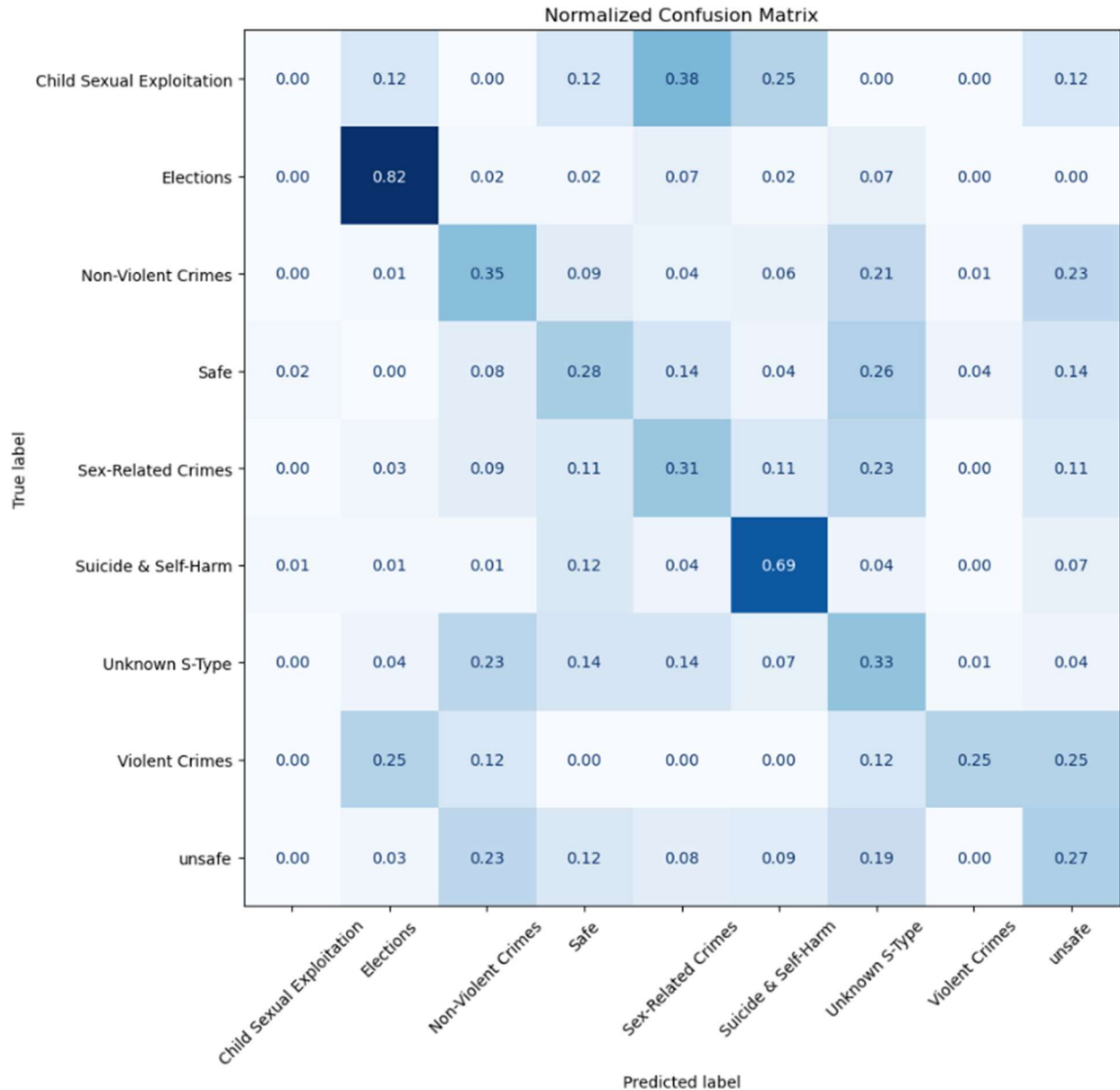
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RNN Results

Accuracy: 0.4220532319391635				
	precision	recall	f1-score	support
Child Sexual Exploitation	0.00	0.00	0.00	8
Elections	0.81	0.82	0.81	61
Non-Violent Crimes	0.53	0.35	0.42	141
Safe	0.23	0.28	0.25	50
Sex-Related Crimes	0.22	0.31	0.26	35
Suicide & Self-Harm	0.64	0.69	0.67	75
Unknown S-Type	0.25	0.33	0.28	73
Violent Crimes	0.33	0.25	0.29	8
unsafe	0.27	0.27	0.27	75
accuracy			0.42	526
macro avg	0.36	0.37	0.36	526
weighted avg	0.44	0.42	0.42	526

From the classification report, there is a large difference in the performance of the classes:

- Elections had the best performance with an F1-score of 0.81 because the model learned language properties for elections well. Similarly, Suicide & Self-Harm achieved an F1-score of 0.67, meaning that the RNN was able to identify expressions of suicidal or self-harming behaviors.
- The performance of Non-Violent Crimes was moderate with an F1-score of 0.42.
- The remaining classes were not performing well as they had an F1-score below 0.30.
- Child Sexual Exploitation had the worst performance because the model had no precision and no recall values. This is due to the very few instances of training data in this category (even after some data augmentation)



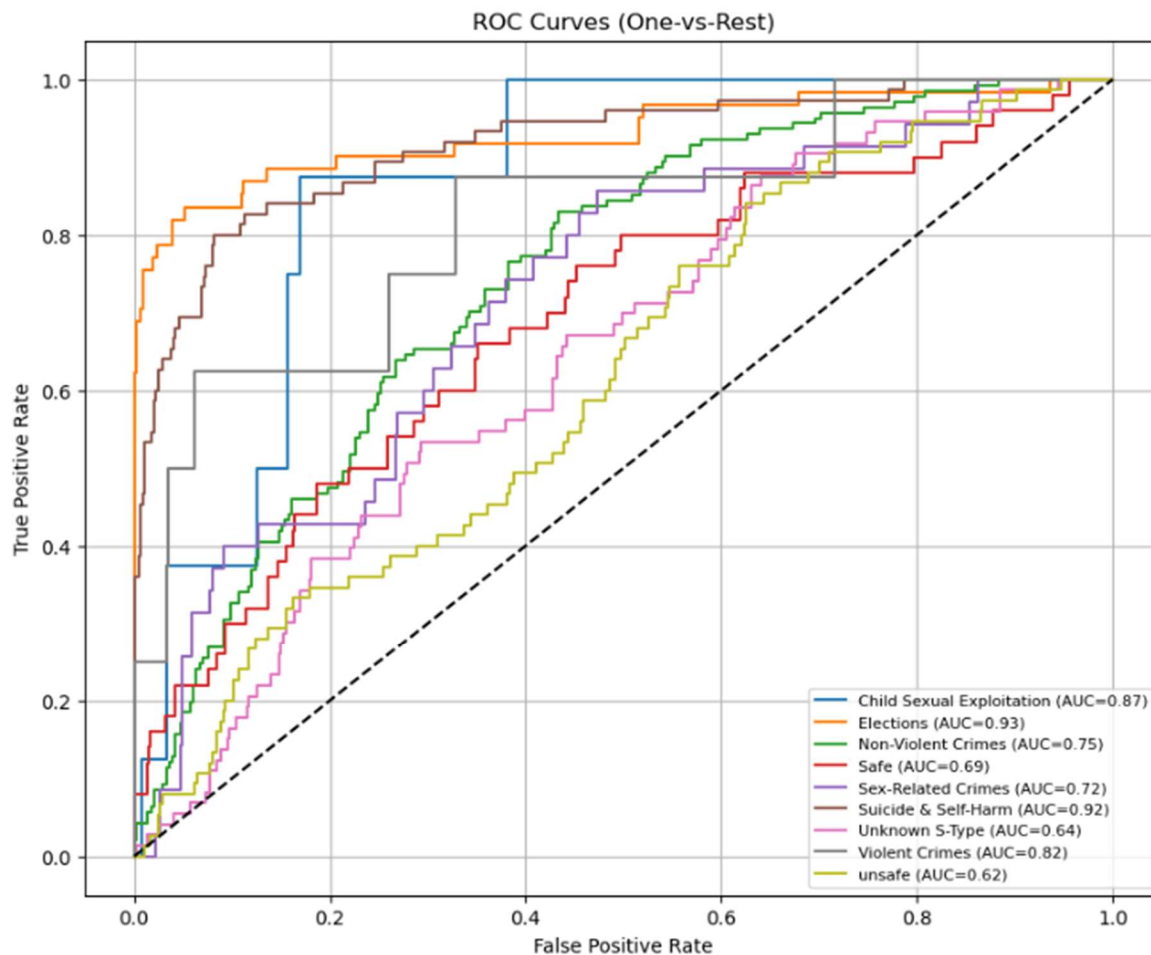
From the confusion matrix, the RNN performed well in classifying many instances from *Election* and *Suicide & Self Harm* categories, as demonstrated by the high diagonal values for these classes.

There is, however, a lot of confusion between remaining classes. Specifically:

- Unsafe samples were confused with Unknown S-Type.
- Safe samples were often wrongly labeled as unsafe classes.
- Non-Violent Crime samples had a lot of overlap with the general unsafe class, which is technically correct but the model needs to be more specific in its classification.

- Minority classes like Violent Crimes and Child Sexual Exploitation showed very few correct classifications.

This suggests that the RNN failed to learn highly discriminative features from minority classes.



AUC may seem unusually high for the minority classes but that is because ROC-AUC measures how well the model ranks examples, not whether it predicts the correct class.

For example, Child Sexual Exploitation class has a high ROC-AUC (0.87) with zero recall, this could mean the model **learned useful information about the class**, but it's not assigning it the highest score often enough.

Macro ROC-AUC : 0.7730

Weighted ROC-AUC : 0.7557

LSTM Results

Accuracy: 0.5247148288973384				
	precision	recall	f1-score	support
Child Sexual Exploitation	0.83	0.62	0.71	8
Elections	0.85	0.77	0.81	61
Non-Violent Crimes	0.57	0.54	0.55	141
Safe	0.29	0.34	0.31	50
Sex-Related Crimes	0.48	0.40	0.44	35
Suicide & Self-Harm	0.73	0.80	0.76	75
Unknown S-Type	0.34	0.37	0.36	73
Violent Crimes	0.78	0.88	0.82	8
unsafe	0.32	0.31	0.31	75
accuracy			0.52	526
macro avg	0.58	0.56	0.56	526
weighted avg	0.53	0.52	0.53	526

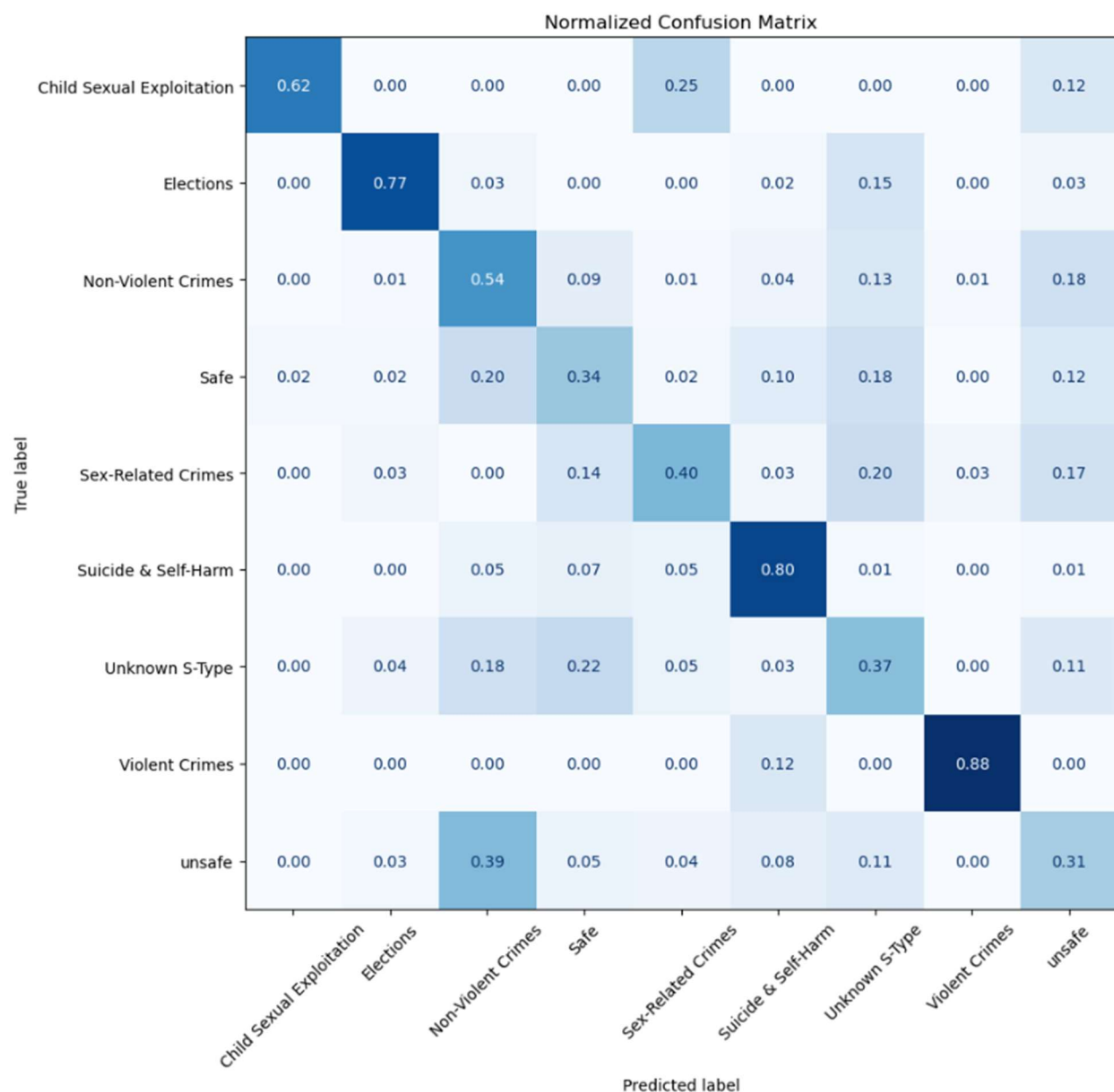
The LSTM network performed better than the RNN in almost all the metrics used.

Elections scored an F1-score of 0.81, and this proves that elections were one of the easiest subjects to classify.

But Improvements were mainly in minority subjects:

- Child Sexual Exploitation greatly improved from an F1-score of 0.00 to 0.71.
- Violent Crimes increased from 0.29 to 0.82.
- Sex Related Crimes improved from 0.26 to 0.44.
- Non Violent Crimes improved from 0.42 to 0.55.
- Suicide & Self-Harm increased from 0.67 to 0.76.

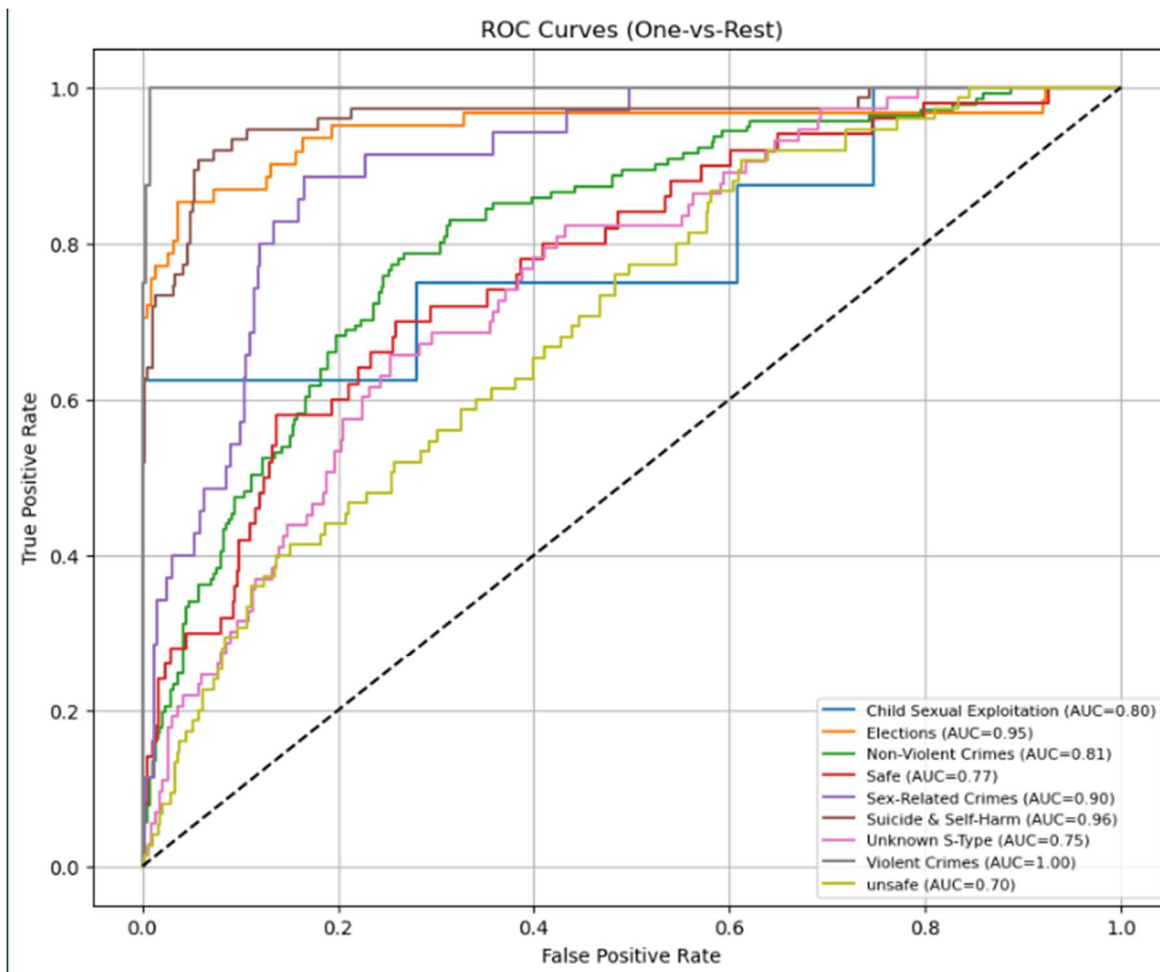
Safe, Unknown S-Type, and Unsafe, although quite difficult to classify, showed some improvement too.



There is a clearer diagonal in the confusion matrix for LSTM than in the case of RNN, implying a higher number of correctly classified observations.

In particular, the model has shown better differentiation among related types of toxicity and also has managed to minimize the confusion in identifying Violent Crimes, Child Sexual Exploitation, and Sex-related crimes.

Though the model could still confuse between safe, unknown S-type, and unsafe, there is still less classification error, implying that LSTM had learned better contexts than the plain RNN.



Macro ROC-AUC : 0.8491

Weighted ROC-AUC : 0.8296

Conclusion:

The findings clearly show the superior performance of the LSTM over the traditional RNN on all measures used for the analysis.

The accuracy score for the overall categorization improved from 42.21% for the RNN to 52.47% for the LSTM model, which is a gain of about 10.3 percent. Also, the same trend was observed for macro-average and weighted-average F1-scores, implying that the LSTM exhibited relatively balanced classification performance on majority and minority classes.

The largest improvement was seen for *Child Sexual Exploitation and Violent Crimes* since the RNN failed to find proper decision boundaries owing to the few samples available for the learning process. On the other hand, the LSTM managed to learn discriminative context features for these classes, which resulted in the improved precision, recall, and F1-score.

Both models showed satisfactory performance on *Election and Suicide & Self-Harm* categories, implying that these classes contain distinct linguistic properties that make their recognition relatively simple.

However, both models continue to struggle with discrimination of Safe, Unknown S-Type, and Unsafe classes since there is more ambiguousness between them.